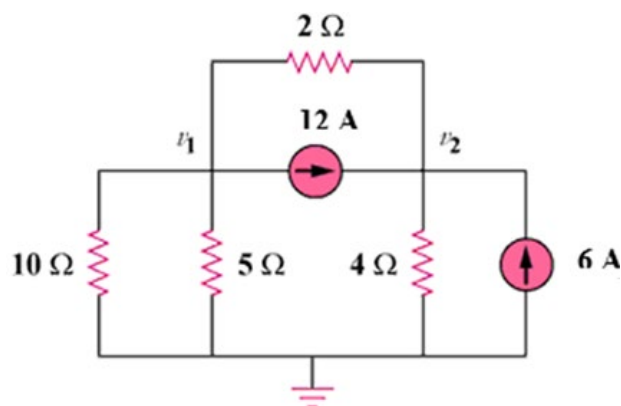
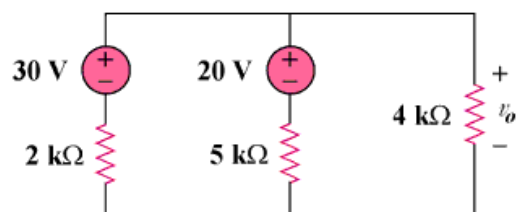


HW 3_Ch3

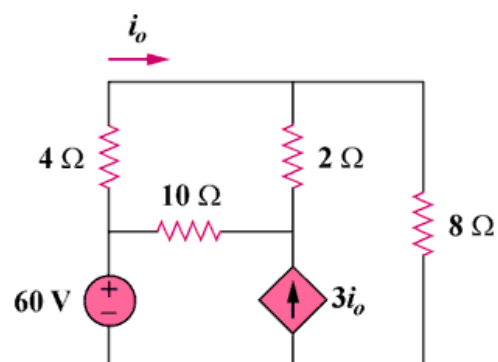
- 1) For the circuit below obtain v_1 and v_2 .



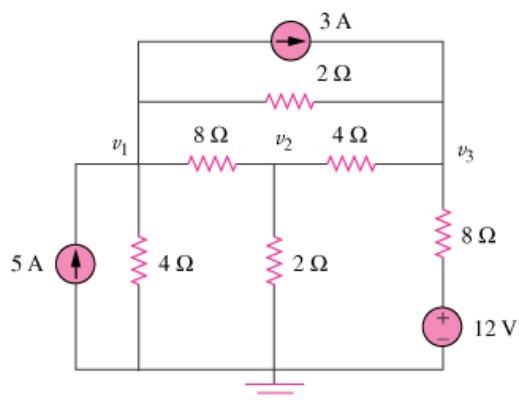
- 2) Obtain v_o in the circuit below



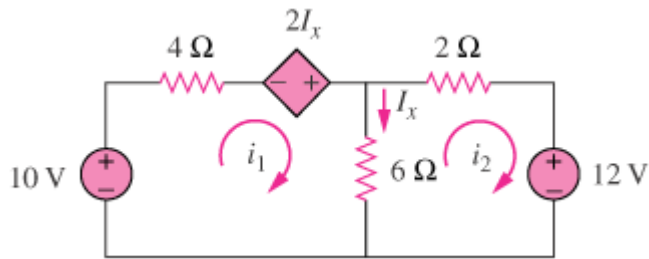
- 3) Using nodal analysis, find current i_o in the circuit below.



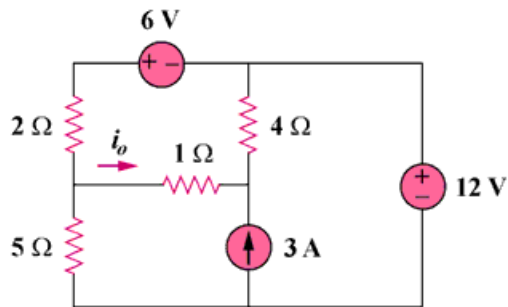
- 4) Use nodal analysis to find v_1 , v_2 , and v_3 in the circuit below.



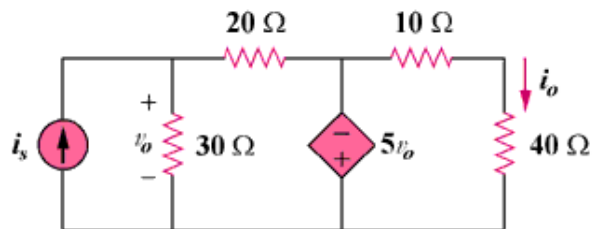
- 5) An audio amplifier with resistance 9Ω supplies power to a speaker. In order that maximum power is delivered, what should be the resistance of the speaker?
- 6) Determine the mesh currents i_1 and i_2 in the circuit below.



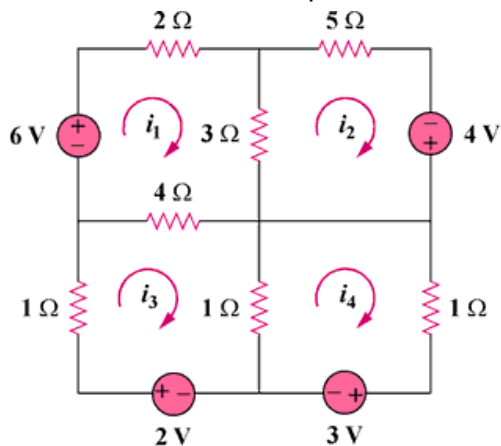
- 7) Use mesh analysis to obtain i_o in the circuit below.



- 8) Calculate the current gain i_o/i_s in the circuit below.



- 9) Write the mesh-current equations for the circuit below.



- 10) For the circuit below, find the gain v_o/v_s .

